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Combinatorics

An Original Olympiad Practice Set

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Original content. These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. Eight original olympiad-flavoured combinatorics problems spanning counting, bijections, the pigeonhole principle, extremal set theory, invariants, and combinatorial games. Each rewards a structural insight rather than brute computation, and every numeric answer has been independently verified by exhaustive search. Every closed-form answer was checked symbolically/numerically. Free to share for study.

Problems

1. DIFFICULTY 2/5 · COUNTING, GAP METHOD, BINOMIAL COEFFICIENTS

A row of 12 lockers is to be painted, exactly 4 of them red and the rest grey, so that no two red lockers are adjacent. In how many ways can this be done?

2. DIFFICULTY 3/5 · GENERATING FUNCTIONS, ROOTS OF UNITY FILTER, COUNTING

Let $S = \{1, 2, 3, \dots, 12\}$. Counting the empty set (whose sum is 0), how many subsets of S have a sum divisible by 3?

3. DIFFICULTY 2/5 · LATTICE PATHS, BIJECTION, BINOMIAL COEFFICIENTS

An ant starts at the bottom-left corner of a 6×4 city grid and walks to the top-right corner, moving only one block East or one block North at a time. How many distinct shortest routes are there?

4. DIFFICULTY 3/5 · INCLUSION-EXCLUSION, SURJECTIONS, COUNTING

Seven distinct prizes are to be handed out to three children named A, B, C, with each child receiving at least one prize. In how many ways can the prizes be distributed?

5. DIFFICULTY 3/5 · BIJECTION, DYCK PATHS, CATALAN NUMBERS, REFLECTION PRINCIPLE

A frog starts at 0 on the number line and makes 12 jumps, each of size 1 either right (+1) or left (−1). It must finish back at 0 and must never be at a negative coordinate at any moment. How many such sequences of 12 jumps are possible?

6. DIFFICULTY 3/5 · INVARIANT, PARITY, MONOVARIENT

The numbers $1, 2, 3, \dots, 2025$ are written on a board. Repeatedly, one erases any two numbers a, b and writes their absolute difference $|a - b|$ in their place, reducing the count by one each time. After 2024 such operations a single number remains. Prove that this final number cannot be 0, and determine whether it must be odd or even.

7. DIFFICULTY 4/5 · COMBINATORIAL GAMES, P-POSITIONS, INDUCTION, MODULAR STRUCTURE

Two players alternately remove stones from a single pile, the player who removes the last stone wins. On each turn a player must remove exactly 1, 3, or 4 stones. Starting from a pile of n stones with optimal play, determine exactly the values of n for which the player who moves first loses, and prove your characterization.

8. DIFFICULTY 4/5 · PIGEONHOLE, EXTREMAL, TIGHTNESS CONSTRUCTION

Let $n \geq 1$. Prove that any $n + 1$ integers chosen from $\{1, 2, 3, \dots, 2n\}$ must contain two distinct numbers that are coprime, and show that the bound $n + 1$ is best possible (i.e. there is a choice of n numbers with no two coprime).

Solutions

1. Answer: 126

Place the 8 grey lockers first; they create 9 gaps (including the two ends) into which red lockers may go:

$$_ G _ G _ G _ G _ G _ G _ G _.$$

Choosing 4 of these 9 gaps to receive a red locker guarantees that no two reds are adjacent, and every valid arrangement arises this way exactly once. Hence the count is

$$\binom{9}{4} = 126.$$

In general, the number of binary strings of length n with exactly k ones and no two ones adjacent is $\binom{n-k+1}{k}$; here $\binom{12-4+1}{4} = \binom{9}{4} = 126$.

Second method. Bijection: map a valid arrangement, with red positions $\{r_1, \dots, r_4\}$

2. Answer: 1376

The total over all subsets is encoded by $\prod_{k=1}^{12} (1 + x^k)$; we want the sum of coefficients of x^m with $3 \mid m$. Apply the roots-of-unity filter with $\omega = e^{2\pi i/3}$: the desired number is

$$N = \frac{1}{3} \sum_{j=0}^2 \prod_{k=1}^{12} (1 + \omega^{jk}).$$

For $j = 0$ the product is $2^{12} = 4096$. For $j = 1$, group the exponents in consecutive triples; over each triple $\{3t + 1, 3t + 2, 3t + 3\}$ we get

$$(1 + \omega)(1 + \omega^2)(1 + \omega^3) = (1 + \omega)(1 + \omega^2) \cdot 2.$$

Since $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$, $(1 + \omega)(1 + \omega^2) = 1 + \omega + \omega^2 + \omega^3 = 1$, so each triple contributes 2. With $12/3 = 4$ triples the product is $2^4 = 16$. The $j = 2$ product is the complex conjugate, also 16. Hence

$$N = \frac{4096 + 16 + 16}{3} = \frac{4128}{3} = 1376.$$

Second method. Residue bookkeeping. The set S splits into residue classes mod 3: $\{3, 6, 9, 12\}$ (residue 0), $\{1, 4, 7, 10\}$ (residue 1), $\{2, 5, 8, 11\}$ (residue 2), four elements each. The four residue-0 elements may be included freely, contributing a factor $2^4 = 16$. For the residue-1 class, if we pick k_1 of its elements the sum contributes $k_1 \pmod{3}$; for the residue-2 class, picking k_2 elements contributes $2k_2 \pmod{3}$. The total sum is divisible by 3 exactly when $k_1 + 2k_2 \equiv 0 \pmod{3}$. Summing $\binom{4}{k_1} \binom{4}{k_2}$ over all $0 \leq k_1, k_2 \leq 4$ with $k_1 + 2k_2 \equiv 0 \pmod{3}$ gives 86 such weighted choices. Multiplying by the free factor: $16 \cdot 86 = 1376$.

3. Answer: 210

A shortest route uses exactly 6 East steps and 4 North steps, in some order, for a total of 10 steps. A route is determined precisely by which 4 of the 10 step-slots are North (equivalently which 6 are East). Therefore the number of routes is

$$\binom{10}{4} = \binom{10}{6} = 210.$$

The key insight is that each monotone path corresponds bijectively to a word in $\{E, N\}$ with 6 E's and 4 N's, so counting paths is counting such words.

Second method. Build a Pascal-style table: write at each lattice point the number of monotone paths reaching it from the start, where each interior value is the sum of the value below it and the value to its left (boundary cells all equal 1). Filling the 7×5 array of lattice points propagates Pascal's rule and yields 210 at the top-right corner.

4. Answer: 1806

We count surjections from the 7 prizes onto the 3 children. Without the 'at least one' restriction there are $3^7 = 2187$ distributions. By inclusion-exclusion, subtract those missing at least one child:

$$\sum_{i=0}^3 (-1)^i \binom{3}{i} (3-i)^7 = 3^7 - \binom{3}{1} 2^7 + \binom{3}{2} 1^7 - \binom{3}{3} 0^7.$$

This equals $2187 - 3 \cdot 128 + 3 \cdot 1 - 0 = 2187 - 384 + 3 = 1806$. The insight is that 'every child nonempty' is exactly the surjection condition, handled cleanly by inclusion-exclusion over which children are excluded.

Second method. Via Stirling numbers: the number of ways to split 7 distinct prizes into 3 nonempty unlabelled groups is the Stirling number $S(7, 3) = 301$; assigning the three groups to the three named children multiplies by $3! = 6$, giving $301 \cdot 6 = 1806$.

5. Answer: 132

Record each jump as $+1$ or -1 ; a valid trip is a sequence of $2n = 12$ steps (so $n = 6$) with n plus-steps and n minus-steps whose every partial sum is ≥ 0 . These are exactly the Dyck paths of semilength n , counted by the Catalan number

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

For $n = 6$,

$$C_6 = \frac{1}{7} \binom{12}{6} = \frac{924}{7} = 132.$$

The insight is recognising the 'never negative, returns to start' walk as a Dyck path.

Second method. Reflection principle: among the $\binom{12}{6} = 924$ sequences with six $+1$ s and six -1 s, count the bad ones that dip to -1 . Reflecting the path after its first visit to -1 bijects the bad sequences with all sequences having 7 minus-steps and 5 plus-steps, of which there are $\binom{12}{5} = 792$. Hence valid $= 924 - 792 = 132$.

6. Answer: \text{The final number is always odd; in particular it cannot be } 0.

(\text{Proof})

Consider the parity of the sum of all numbers currently on the board. When a and b are replaced by $|a - b|$, the sum changes by

$$|a - b| - (a + b).$$

Since $|a - b|$ and $a + b$ have the same parity (they differ by $2 \min(a, b)$, an even number), the sum's parity is unchanged by every operation. Thus the parity of the total is an invariant. Initially the total is

$$1 + 2 + \cdots + 2025 = \frac{2025 \cdot 2026}{2} = 2025 \cdot 1013 = 2051325,$$

which is odd. Therefore at every stage the sum is odd, and in particular the final lone number equals that odd sum and is odd. An odd number is never 0, so 0 can never be the result. ■

Second method. Work modulo 2 throughout: replacing a, b by $|a - b| \equiv a + b \pmod{2}$ is, mod 2, the same as replacing two bits by their sum, which preserves the total mod 2. The number of odd values among $1, \dots, 2025$ is 1013, which is odd, so the parity of the sum is 1; this parity is conserved, forcing an odd final value.

7. Answer: \text{First player loses exactly when } n \equiv 0 \text{ or } 2 \pmod{7}.

(\text{Proof})

Call a position a P-position (previous player wins, i.e. the player to move loses) or an N-position (next player to move wins). We claim n is a P-position iff $n \equiv 0$ or $2 \pmod{7}$. Proof by strong induction on n , using two facts: a position is P iff every move leads to an N-position; it is N iff some move leads to a P-position.

Base cases ($0 \leq n \leq 6$): $n = 0$ is a P-position (no move; the player to move has already lost). $n = 1, 3, 4$ are N (remove all stones and win). $n = 2$: the only legal moves are to remove 1 (leaving 1, an N-position) — 3, 4 are too big — so every move leads to N, making 2 a P-position. $n = 5$: remove 3 to leave 2 (P), so N. $n = 6$: remove 4 to leave 2 (P), so N. This matches the claim on $0, \dots, 6$: P exactly at $0, 2$.

Inductive step ($n \geq 7$): suppose the claim holds for all smaller values. The positions reachable from n are $n - 1, n - 3, n - 4$. Compute residues mod 7:

- If $n \equiv 0$: the moves reach residues 6, 4, 3, none of which is 0 or 2; so all moves land in N-positions, hence n is a P-position. ✓
- If $n \equiv 2$: moves reach residues 1, 6, 5, again none equal to 0 or 2; all land in N, so n is P. ✓
- If $n \equiv 1$: removing 1 reaches residue 0 (a P-position), so n is N.
- If $n \equiv 3$: removing 1 reaches residue 2 (P), so N.
- If $n \equiv 4$: removing 4 reaches residue 0 (P), so N.
- If $n \equiv 5$: removing 3 reaches residue 2 (P), so N.
- If $n \equiv 6$: removing 4 reaches residue 2 (P), so N.

In every non- $\{0, 2\}$ residue there is a move into a P-position, so those are N. This completes the induction: the first player loses precisely when $n \equiv 0$ or $2 \pmod{7}$. ■

Second method. Sprague–Grundy: define $g(n) = \text{mex}\{g(n - 1), g(n - 3), g(n - 4)\}$ where moves to negative piles are unavailable. Computing g shows it is periodic with period 7, and the zero values (losing positions) occur exactly at $n \equiv 0, 2 \pmod{7}$; a position is losing iff its Grundy value is 0.

8. Answer: \text{Proof; the bound } n+1 \text{ is sharp.}

Partition $\{1, 2, \dots, 2n\}$ into the n consecutive pairs

$$\{1, 2\}, \{3, 4\}, \{5, 6\}, \dots, \{2n-1, 2n\}.$$

If we select $n+1$ numbers, then by the pigeonhole principle ($n+1$ objects into n boxes) two of the chosen numbers fall in the same pair $\{2k-1, 2k\}$. Two consecutive integers $2k-1$ and $2k$ are always coprime, since any common divisor of them divides their difference 1; hence $\gcd(2k-1, 2k) = 1$. Thus the selection contains two coprime numbers.

Sharpness: the set of the n even numbers $\{2, 4, 6, \dots, 2n\} \subseteq \{1, \dots, 2n\}$ has size n , and any two of its elements share the common factor 2, so no two are coprime. Therefore n numbers need not contain a coprime pair, and the threshold $n+1$ cannot be lowered. ■

Second method. Alternative tight construction and counting view: one might try to take n numbers all sharing a fixed prime divisor; in $\{1, \dots, 2n\}$ the prime 2 furnishes exactly n multiples $\{2, 4, \dots, 2n\}$, the largest such pairwise-non-coprime block. Any prime $p \geq 3$ has only $\lfloor 2n/p \rfloor$